

SAUHARDHA KAFLE

Computer Engineer

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ABOUT ME

I'm Sauhardha Kafle, a Computer Engineer and developer passionate about building clean, efficient, and user-friendly applications. I enjoy working across the stack, from designing frontend interfaces, backend system to writing Low-level and System level programming. I also play with Deep Learning, Natural Language Processing models.

SKILLS

Languages: Rust, TypeScript, Python
Backend: Actix-web, NodeJS, ExpressJS, AIOHTTP, Flask, FastAPI
Frontend: NextJS, ReactJS, TailwindCSS,
Databases & Tools: Redis, MySQL, MongoDB, Appwrite
AI/ML: Deep Learning, Natural Language Processing
Infra: Docker, gRPC, IPC, webRTC, websocket

PROJECTS

[Torrex](#) | *Rust*

- A bittorrent client implementation from scratch in **rust** programming language.
- Feature includes, downloading data using torrent file or magnet link that supports **TCP connection**.

[nutshell](#) | *TypeScript, Rust, NextJS*

- A developer platform to build and deploy web application on cloud.
- Uses **Cloudflare's R2 storage**, **docker** to build the code, **redis** for information flow, **Rust** to handle the request, **NextJS** for frontend and **prisma postgres** for database.

[CodeRoom](#) | *NodeJS, MySQL, Docker*

- A **real-time** collaborative coding platform with multi-user editing, chat, and live code compilation.
- Uses **docker** to compile code, **web socket** for real time messaging in **NodeJs**. Other feature includes **Authentication**, **OTP** code and uses **MySQL** for database.

[dln-rust](#) | *Rust*

- A TUI-based asynchronous download manager leveraging IPC (Unix Domain Socket) for inter-process communication.
- Built with **ratatui** for the terminal interface and **tokio** for async runtime. Features concurrent multi-link downloads, pause/resume functionality, and async task monitoring with **tokio-console**.

[Pneumonia Detection](#) | *TensorFlow, Keras*

- Implemented a multi-layer **CNN** in **TensorFlow/Keras** to classify pneumonia from chest X-ray images.
- Includes data preprocessing, training, and evaluation, achieving high recall (~98%) and strong overall accuracy.

[Aether](#) | *Python, Rust, WebRTC, Websocket*

- A browser-based remote desktop sharing platform. A end-to-end system for screen capturing and streaming using **webRTC**.
- Backend coordination server manages session signaling, peer connections, and room state video transfer and reflecting mouse pointer.

EDUCATION

Kathmandu University

Bachelor in Computer Engineer

2022 – 2027

EXPERIENCE

Volunteer

- KURC volunteer – 2023
- KUCC volunteer – Nov 2023

Events

- CS50 webinar organizer
- Idea pitching event manager
- **ITMeet-2025 Technical Lead**